pkt
LOG

Architecture & Technical Reference

v1.0 · Internal Documentation

System Components

Ingest Engine

The ingest engine binds dual sockets (UDP and TCP) on the configured syslog port (default 8761). It runs a continuous receive loop that feeds parsed events through a normalize-and-batch pipeline before writing to ClickHouse.

Syslog Parser

Auto-detects RFC 3164 (BSD syslog) vs RFC 5424 (structured syslog) by inspecting the PRI header. Extracts priority, facility, severity, timestamp, hostname, app_name, pid, message body, and RFC 5424 structured data elements.

Normalizer

Maps numeric priority to human-readable severity (emergency → debug) and facility (kern, auth, daemon, etc.) strings. Applies site/group/collector enrichment by looking up the sender IP in the collector registry. Adds a server-side `received_at` timestamp.

Batch Buffer

Accumulates normalized records in memory. Flushes to ClickHouse when the flush interval elapses (default 5 seconds) or the maximum batch size is reached. Thread-safe.

Ingest Journal

If ClickHouse is unavailable at flush time, the batch is written to a local journal file. The journal replayer monitors ClickHouse availability and resubmits journal entries when storage recovers.

ClickHouse Schema

```
CREATE TABLE pktlog.syslog_events (
    received_at     DateTime,            -- server ingest timestamp
    event_time      DateTime,            -- parsed message timestamp
    host            String,              -- source hostname / IP
    severity        LowCardinality(String),
    facility        LowCardinality(String),
    priority        UInt8,               -- raw PRI (0-191)
    program         String,              -- process / app name
    pid             String,
    message         String,              -- parsed message body
    raw_message     String,              -- original syslog line
    collector_ip    String,
    collector_name  String,
    site            LowCardinality(String),
    group_name      LowCardinality(String),
    msg_id          String,              -- RFC 5424 MSGID
    structured_data String,              -- RFC 5424 SD element
    ingest_id       UUID DEFAULT generateUUIDv4()
) ENGINE = MergeTree()
  PARTITION BY toYYYYMM(received_at)
  ORDER BY (received_at, host, severity)
  TTL received_at + INTERVAL 90 DAY;
```

SQLite Tables

Table	Contents
`users`	username, hashed_password, role, is_active
`settings`	key/value app configuration
`alert_rules`	rule type, parameters, enabled flag
`alert_events`	fired events, status, timestamps
`collector_registry`	IP, name, site, group, last_seen

API Endpoints

Ingest

Method	Path	Auth	Description
POST	/api/ingest/syslog	Bearer token	Accept JSON batch from collectors
GET	/api/ingest/stats	JWT	Buffer and journal statistics

Syslog Query

Method	Path	Description
GET	/api/syslog/search	Paginated search with filters
GET	/api/syslog/stats	Aggregate counts by severity/facility
GET	/api/syslog/timeseries	Volume per time bucket
GET	/api/syslog/export	Download as CSV or JSON
GET	/api/syslog/top-hosts	Top N hosts by event count

Alerts

Method	Path	Auth	Description
GET	/api/alerts/rules	JWT	List rules
POST	/api/alerts/rules	Admin	Create rule
DELETE	/api/alerts/rules/{id}	Admin	Delete rule
GET	/api/alerts/events	JWT	List fired events

System

Method	Path	Auth	Description
GET	/api/health	None	Service health check
POST	/api/system/restart	Admin	Restart service
POST	/api/system/backup	Admin	Trigger backup

Ingest Pipeline — Step by Step

Step 1 — Receive
UDP datagrams or TCP stream messages arrive on port 8761 and are queued for parsing.

Step 2 — Parse
RFC format is auto-detected. Syslog PRI header is decoded to severity (bits 0–2) and facility (bits 3–7). Timestamps are normalized to UTC.

Step 3 — Normalize
Numeric codes become human-readable strings. Collector registry lookup adds `collector_name` , `site` , and `group_name` . Server timestamp is attached.

Step 4 — Buffer
Records accumulate in the in-memory batch buffer until the flush interval (5s) or max size is reached.

Step 5 — Write
ClickHouse bulk INSERT. On failure: journal write. Journal replayer retries on recovery.

Deployment Requirements

Component	Requirement
Python	3.9+ (3.11+ recommended)
ClickHouse	24.x+
Node.js	18+ (frontend build only)
OS	Amazon Linux 2023 / RHEL 8+ / Ubuntu 22+
Process manager	systemd

Ports used:

- `8768` — HTTPS web interface and REST API
- `8761` — Syslog ingest (UDP + TCP)
- `9000` — ClickHouse native protocol (localhost only)

Directory Layout (Server)

```
/app-dir/
├── app/             FastAPI application
├── frontend/dist/  Built React assets (served by FastAPI)
├── ingest_journal/ Fallback journal files
├── pktlog.db       SQLite database
├── config.yaml     Runtime configuration
└── pktlog.service  systemd unit
```